

# TrES-3b

R-0343

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-3\_230602 · created 2026-07-18T07:26:33+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460097.7123 +/- 0.0017 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.196 +/- 0.052                 |
| Transit depth (Rp/Rs)^2                           | 3.84 +/- 2.02 [%]               |
| Orbital Inclination (inc)                         | 86.82 +/- 2.47                  |
| Airmass coefficient 1 (a1)                        | 0.991 +/- 0.094                 |
| Airmass coefficient 2 (a2)                        | 0.008 +/- 0.041                 |
| Scatter in the residuals of the lightcurve fit is | 9.38 %                          |
| Best Comparison Star                              | #2 - [214, 263]                 |
| Optimal Aperture                                  | 2.52                            |
| Optimal Annulus                                   | 6.84                            |
| Transit Duration (day)                            | 0.0805 +/- 0.0092               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                         |
|--------------------------------------|---------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.196 $\pm$ 0.052 vs 0.16309 (deviation 0.57 $\sigma$ ) |
| Duration fit vs published            | 0.0805 d vs 0.0567 d (deviation 2.33 $\sigma$ )         |
| Mid-time O–C                         | 2.1 min                                                 |
| Depth SNR                            | 3.4                                                     |
| Residual scatter                     | 9.38 %                                                  |

## Light curve

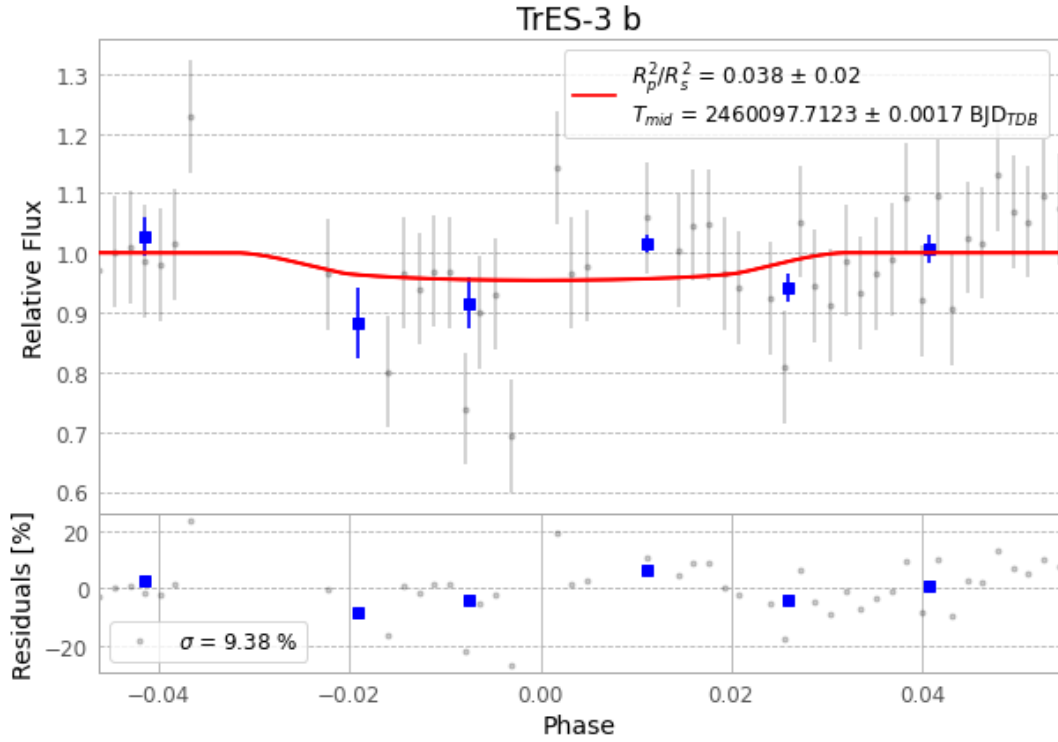


Figure 1 — FinalLightCurve\_TrES-3 b\_2023-06-01.png (SHA-256 f92897eb19370804...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 178], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 fdce9dcc9317c68f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 8b944da

## Data provenance

64 FITS frames (42 MB), TRES-3230602033315.FITS → TRES-3230602064215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230602.log (0ad67cdf5b54...), FinalLightCurve\_TrES-3 b\_2023-06-01.png (f92897eb1937...), FinalLightCurve\_TrES-3 b\_2023-06-01.csv (8a3e02546e5b...)

## Compute resources

Wall time 170.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 07:26Z.